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AUSHADH : Smart Medicine Box

A Smart Healthcare Device for Medication Adherence Monitoring

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Abstract / Project Summary

This project presents the design and development of an IoT-Based Smart Medicine Box to help patients — especially elderly individuals and those with chronic conditions — manage their medication schedules reliably. The main objective is to provide real confirmation of pill consumption rather than merely issuing reminders, thereby addressing the critical challenge of medication non-adherence.

The device is built using an ESP32 microcontroller as the central processing unit, along with FSR (Force Sensitive Resistor) / load cell sensors under each compartment to detect actual weight change when a pill is removed. A 4×4 matrix keypad allows schedule configuration, an LCD display shows real-time status, a buzzer provides audible reminders, and per-compartment LED indicators give visual feedback (Green = Taken, Red = Missed).

When a dose is due, the device alerts the patient via buzzer and LED. Once the pill is removed, the weight sensor confirms removal and updates the cloud via Wi-Fi. If no removal is detected within the configured time window, the caregiver or doctor is notified remotely.

The system is designed to be compact, affordable, and privacy-friendly — using weight sensing instead of cameras. Its cloud connectivity through Wi-Fi enables remote monitoring by caregivers and medical professionals from anywhere, making it suitable for home care, assisted living, and clinical settings.

Introduction

Background of the Project

Medication non-adherence is one of the most critical challenges in modern healthcare. Studies estimate that approximately 50% of patients with chronic illnesses fail to take their medications as prescribed, leading to disease progression, preventable hospitalizations, and increased healthcare costs. Elderly patients managing multiple daily medications are at particular risk.

Over the past decade, smart healthcare devices have emerged as a promising solution to bridge the gap between patient intent and actual medical compliance. IoT-enabled devices can monitor, alert, log, and communicate in real time — transforming passive pill boxes into intelligent healthcare assistants.

Problem Context

Existing smart pill dispensers provide reminders through alarms, mobile notifications, or SMS alerts. However, most current systems only remind patients to take medicines but do not confirm whether the medicine has actually been consumed. This gap between reminder and confirmation is the core problem our system solves.

The IoT-Based Smart Medicine Box is a compact, sensor-driven device that integrates weight sensing, real-time alerts, and cloud connectivity to bridge this gap. By placing FSR sensors beneath each pill compartment, the system detects the actual physical removal of medicine — providing true confirmation, not assumption.

The hand-drawn design sketch below shows the initial concept for the horizontal enclosure, with a 4×4 keypad, LCD display, and dual-column compartment layout:

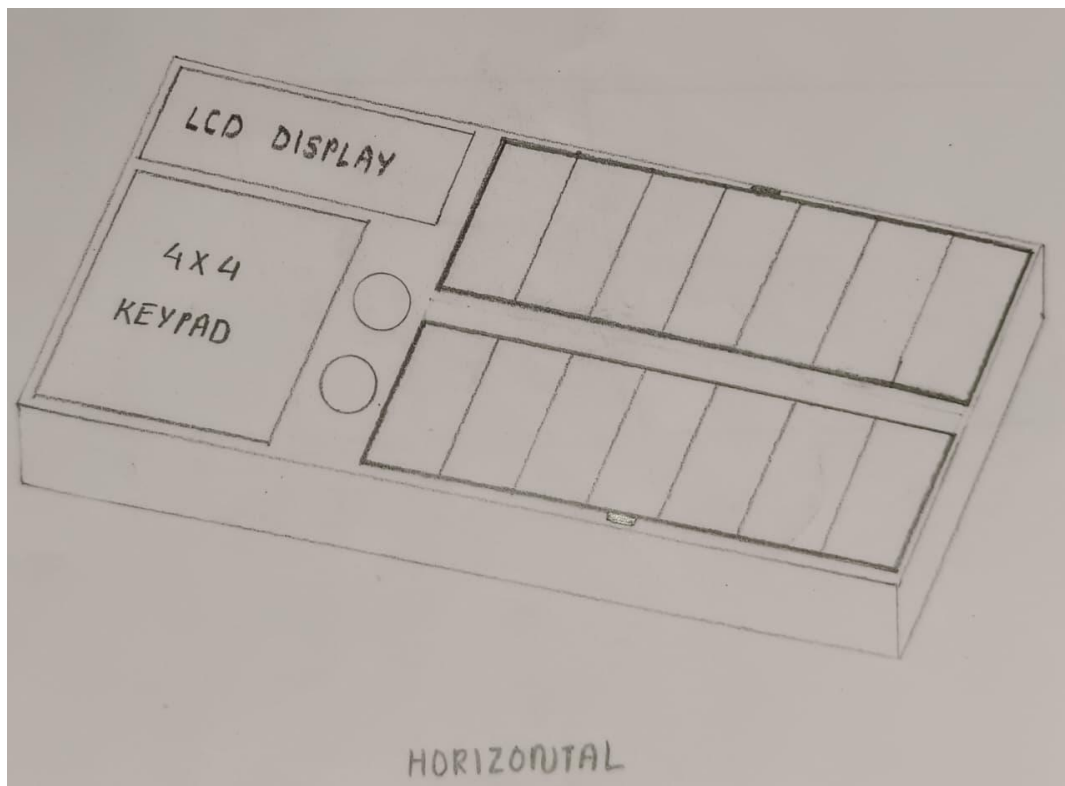


Figure 1: Hand-Drawn Design Sketch of Smart Medicine Box

Problem Statement

Medication non-adherence remains a serious public health issue, particularly among elderly patients, individuals with chronic conditions, and those on complex multi-drug regimens. Despite the availability of basic pill reminders and alarm-based pill boxes, the fundamental problem persists: these systems tell a patient it is time to take medicine, but they have no way of knowing whether the medicine was actually taken.

- Absence of consumption confirmation: A buzzer or phone notification can be dismissed or ignored. Without confirmation, caregivers and doctors cannot distinguish between adherence and non-adherence.
- Heavy reliance on smartphones and internet connectivity: Many elderly patients do not own smartphones or are uncomfortable using apps, making app-based reminders ineffective for a large portion of the target population.
- Privacy concerns with camera-based monitoring: Placing a camera in a patient's private space — particularly in a bedroom or bathroom — is ethically problematic and widely resisted.
- Expensive subscription-based products: Commercial smart dispensers such as MedMinder and Hero Health are priced as expensive subscription-based products, making them inaccessible for lower-income households and developing regions.
- No real-time caregiver integration: Most existing systems lack reliable real-time notification to caregivers when a dose is missed.

Therefore, there is a strong need for a smart, integrated medicine management system that can:

- Confirm actual pill removal through physical sensing
- Alert patients through multiple channels (sound, light, display)
- Notify caregivers remotely via cloud connectivity
- Operate without a camera, preserving patient privacy
- Use affordable, widely available components

Objectives

The main objective of this project is to develop a reliable, affordable, and easy-to-use IoT device that ensures actual medication adherence through real-time sensor-based confirmation and remote monitoring. The specific objectives are:

- Confirm actual pill consumption rather than simply reminding the patient. The FSR weight sensor detects real weight change when a compartment is opened and a pill is removed.
- Enable remote caregiver monitoring through Wi-Fi connectivity. Cloud-based data logging allows doctors, family members, or professional caregivers to check medication adherence status from anywhere in real time.
- Provide multi-modal alerts through a combination of a buzzer (audible), LED indicators per compartment (visual), and LCD display messages (informational), ensuring the patient is alerted through multiple channels simultaneously.
- Ensure patient privacy by using weight sensors instead of cameras, making the system privacy-friendly and suitable for use in bedrooms, bathrooms, or any private space.
- Cost-effective implementation using readily available components (ESP32, HX711, FSR sensors) to create a device affordable enough for home use and scalable for hospital deployment.
- User-friendly design for elderly patients, with large keypad buttons, a clear LCD interface, and simple one-button operation requiring no technical knowledge.
- Develop a 14-compartment weekly tray supporting twice-daily dosing schedules (morning and evening) for all seven days of the week.
- Implement cloud data logging with timestamps to maintain a comprehensive record of taken and missed doses for medical review.

Literature Review / Case Studies

A comprehensive review of existing smart pill dispensers and medication adherence systems was carried out. The comparison below evaluates leading commercial systems against the proposed IoT Smart Medicine Box across key functional parameters.

Table 1: Comparison of Existing Smart Medicine Systems

Feature	MedMinder	Hero Health	AdhereTech	Pill Timer	Our System
Weight-Based Pill Confirmation	X	X	X	X	✓
FSR / Load Cell Sensor	X	X	X	X	✓
LCD Status Display	~	✓	X	X	✓
Audible Buzzer Alert	✓	✓	X	✓	✓
LED Indicators Per Compartment	X	X	X	X	✓
Wi-Fi Cloud Monitoring	✓	✓	✓	X	✓
Weekly 14-Compartment Tray	✓	✓	X	✓	✓
No Camera (Privacy-Friendly)	✓	✓	✓	✓	✓
Caregiver Remote Access	✓	✓	X	X	✓
Keypad Configuration	X	✓	X	X	✓

The comparison demonstrates that the IoT Smart Medicine Box is the only system to incorporate weight-based pill confirmation — the key differentiator that sets it apart from all existing commercial products. While systems like MedMinder and Hero Health offer cloud connectivity, none confirm actual pill removal through physical sensing. Our system achieves this at a fraction of the cost of commercial subscription-based dispensers.

Academic literature in the domain of medication adherence consistently highlights that confirmation-based systems outperform reminder-only systems by 30–40% in long-term adherence rates (WHO, 2003). The integration of IoT with healthcare — specifically through ESP32-class microcontrollers — has been validated in numerous recent publications as a viable, low-cost approach to smart health monitoring.

User Study / Requirement Analysis

The success of a smart healthcare device depends heavily on how well it aligns with the needs, limitations, and behaviour of its target users. For the IoT-Based Smart Medicine Box, a detailed requirement analysis was carried out by considering real-world challenges faced by patients, especially elderly individuals and those with chronic illnesses.

Target User Identification

The primary users of the system include:

- Elderly patients who often forget medication schedules
- Patients with chronic diseases requiring multiple daily doses
- Post-operative patients under strict medication routines
- Caregivers and family members who monitor patient adherence

These users typically require a system that is simple, reliable, and does not depend heavily on technical knowledge.

User Challenges Identified

Based on observations and common healthcare studies, the following key problems were identified:

- Forgetfulness and irregular schedules
- Confusion due to multiple medicines at different times
- Difficulty using smartphone applications (especially among elderly users)
- Lack of confirmation whether medicine was actually taken
- Absence of real-time monitoring for caregivers
- Privacy concerns with camera-based systems

Functional Requirements

Table 2: Functional Requirements Summary

Requirement	Description
Medication Scheduling	Allow users to set daily medicine timings (morning/evening) via keypad
Alert Mechanism	Generate alerts using buzzer (audible), LED indicators (visual), and LCD display
Consumption Confirmation	Detect actual pill removal using FSR/load cell sensors; update status as Taken or Missed
Remote Monitoring	Send real-time notifications to caregivers via Telegram and Blynk cloud dashboard
Data Logging	Maintain records of medication times, taken/missed status, and timestamps

Hardware & Software Requirements

Table 3: Hardware Requirements

Component	Purpose
ESP32 Microcontroller	Central processing unit + Wi-Fi communication

Component	Purpose
FSR / Load Cell Sensors	Weight detection for pill confirmation
HX711 Module	Signal amplification for weight sensors
4×4 Matrix Keypad	User input for scheduling and navigation
16×2 LCD Display (I2C)	Output interface for status and messages
Buzzer	Audible medication alerts
LED Indicators (×14)	Per-compartment visual status (Green/Red)
Power Supply (USB-C)	Device power input

Table 4: Software Requirements

Software / Platform	Purpose
Arduino IDE / Embedded C	Microcontroller programming
Blynk Platform	Cloud dashboard for real-time monitoring
Telegram API	Push notifications to caregivers
NTP Protocol	Internet-based time synchronization

System Architecture

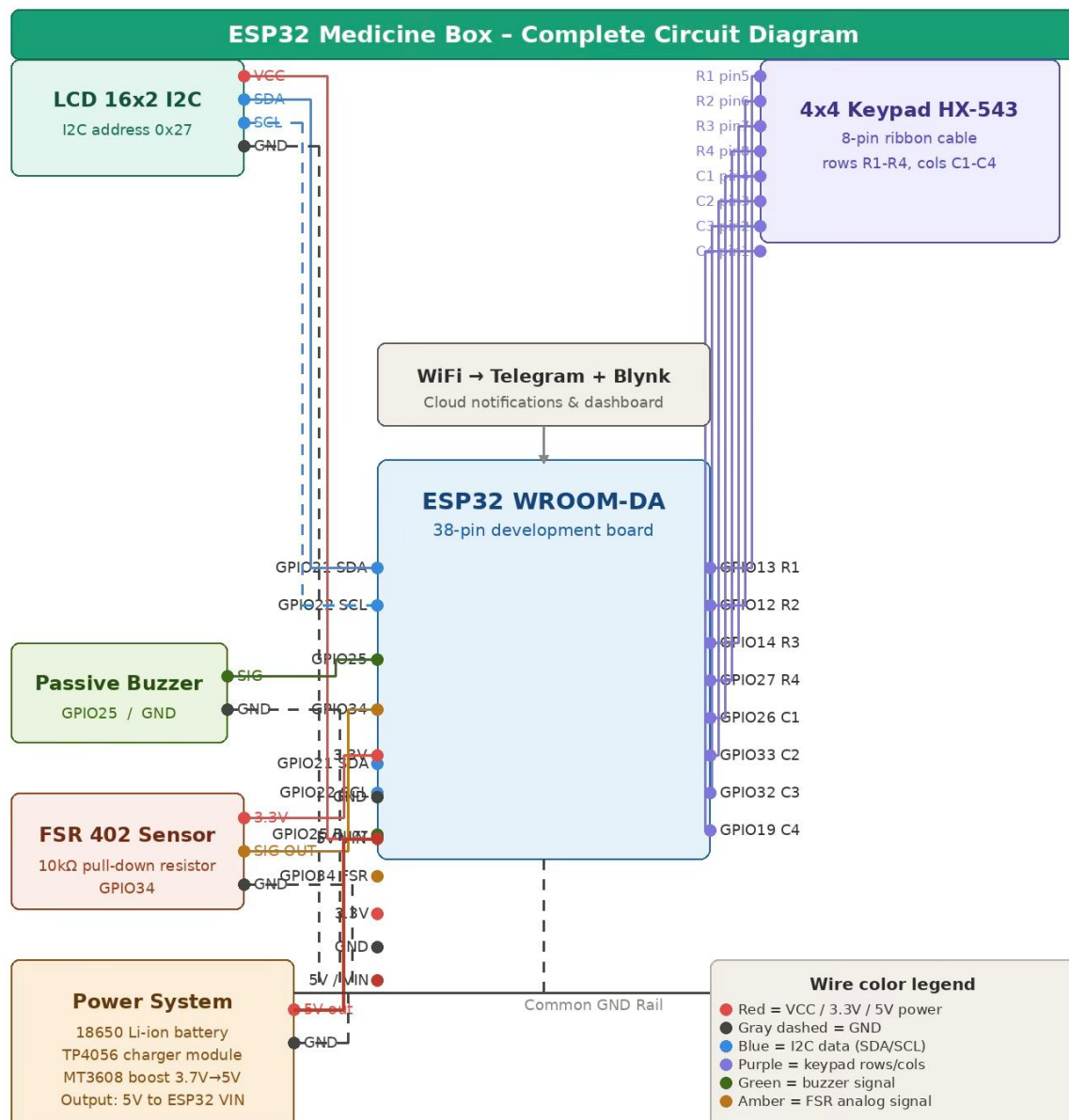


Figure 2: System Architecture / Requirement Analysis Diagram

Non-Functional Requirements

- Usability: Simple interface using keypad and LCD; easy to operate without prior technical knowledge
- Reliability: Accurate detection of pill removal; consistent alarm triggering
- Performance: Real-time response to sensor input; fast notification delivery within a few seconds
- Affordability: Use of low-cost components (ESP32, FSR sensors)
- Privacy: No cameras or intrusive monitoring methods
- Scalability: Extendable to more compartments or advanced monitoring systems

Requirement Justification

- FSR sensors solve the lack of confirmation — the core gap in existing systems
- Keypad interface removes smartphone dependency for elderly users
- Cloud integration enables remote monitoring for caregivers
- Multi-modal alerts improve reliability for users with hearing or visual limitations

Concept Development

Ideation Sketches

The concept development phase began with hand-drawn ideation sketches to rapidly explore different form factors, interaction paradigms, and mechanical configurations. The initial sketches explored the overall enclosure geometry, compartment arrangement, and placement of user-interface elements such as the keypad and LCD display.

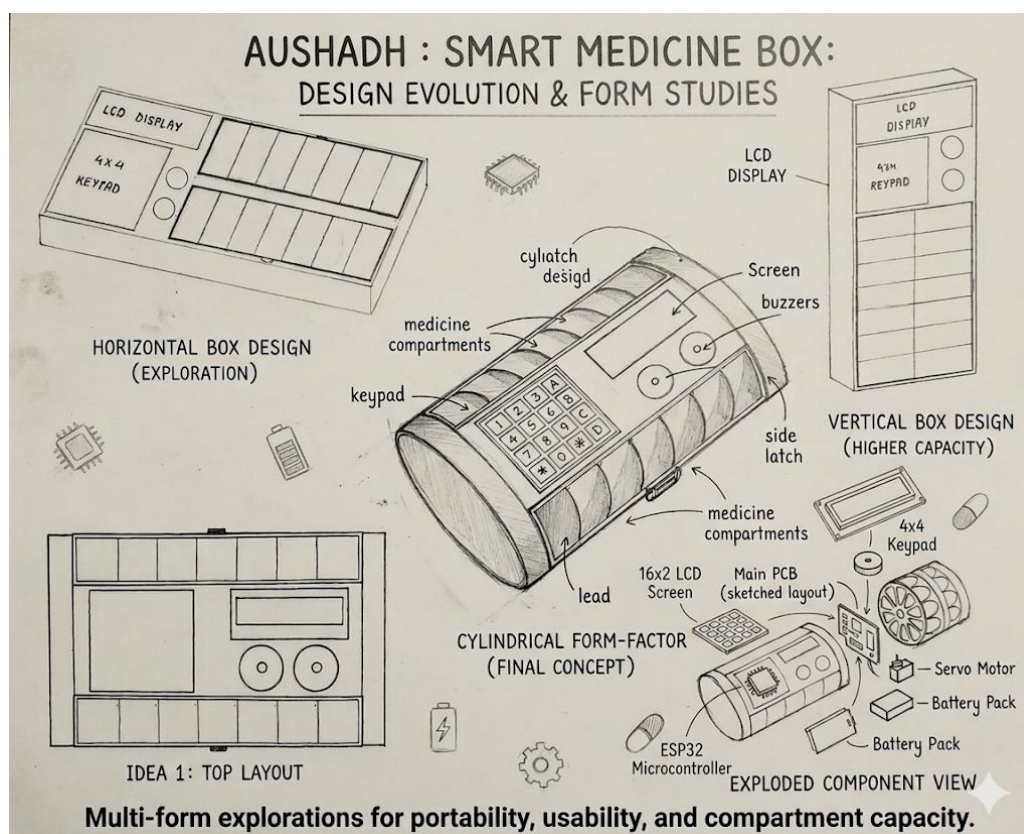


Figure 3: Ideation Concept Sketches – Multiple Concepts Explored

Multiple Concepts Explored

Three primary design concepts were explored during the ideation phase:

Concept A: Vertical Tower Design

A vertical, stackable tower configuration with compartments arranged in a single column. This concept offered minimal footprint but suffered from difficult access to lower compartments and poor visibility of the display.

Concept B: Rotary Carousel Design

A rotating carousel-style dispenser where compartments rotate to a dispensing window at a scheduled time. This concept was mechanically elegant but significantly more complex to manufacture and maintain, with higher risk of mechanical failure.

Concept C: Horizontal Tray Design (Selected)

A flat, horizontal enclosure with two rows of seven compartments arranged on the top panel. The keypad and LCD occupy the front face, while the buzzer grilles and LED indicators are visible from the front. This design prioritises ease of loading and clarity of use.

Concept Selection with Justification

Table 5: Concept Evaluation Matrix

Criteria	Concept A (Tower)	Concept B (Rotary)	Concept C (Horizontal)
Ease of Use	2/5	3/5	5/5
Manufacturing Simplicity	3/5	2/5	5/5
Sensor Integration	3/5	2/5	5/5
Compactness	5/5	3/5	4/5
Cost	4/5	2/5	5/5
Total Score	17/25	12/25	24/25

Concept C — the Horizontal Tray Design — was selected based on the highest overall score. Its flat profile enables straightforward FSR sensor placement under each compartment, simplifies 3D printing, and provides a familiar form factor similar to existing weekly pill organisers, reducing the learning curve for elderly users.

Design Development

Final Design Details

The final design of the IoT Smart Medicine Box is a horizontal, compact, desktop-friendly unit with overall enclosure dimensions of approximately 200mm × 150mm × 100mm, fabricated from white matte ABS/PLA plastic using 3D printing technology. The overall aesthetic is inspired by medical device design standards — smooth matte surfaces, rounded corners, and a neutral white colour — making it non-intimidating and easy to use for elderly patients.

- Top Panel: Houses 14 transparent pill compartments arranged in 2 rows of 7, each with a hinged clear acrylic lid. Each compartment sits above an FSR sensor pad. A small LED indicator is mounted at the front edge of each compartment.
- Front Face: Features the 4×4 keypad on the left and the LCD display on the upper right. Two circular buzzer grilles are mounted between the keypad and compartments.
- Right Side Panel: Contains the USB-C charging port and the power on/off button.
- Internal PCB: The ESP32 board, HX711 modules, and power regulation components are mounted on a custom PCB inside the base of the enclosure. Internal wiring is routed through dedicated channels.

CAD Models / Drawings



Figure 4: CAD Drawing – Medicine Box Assembly

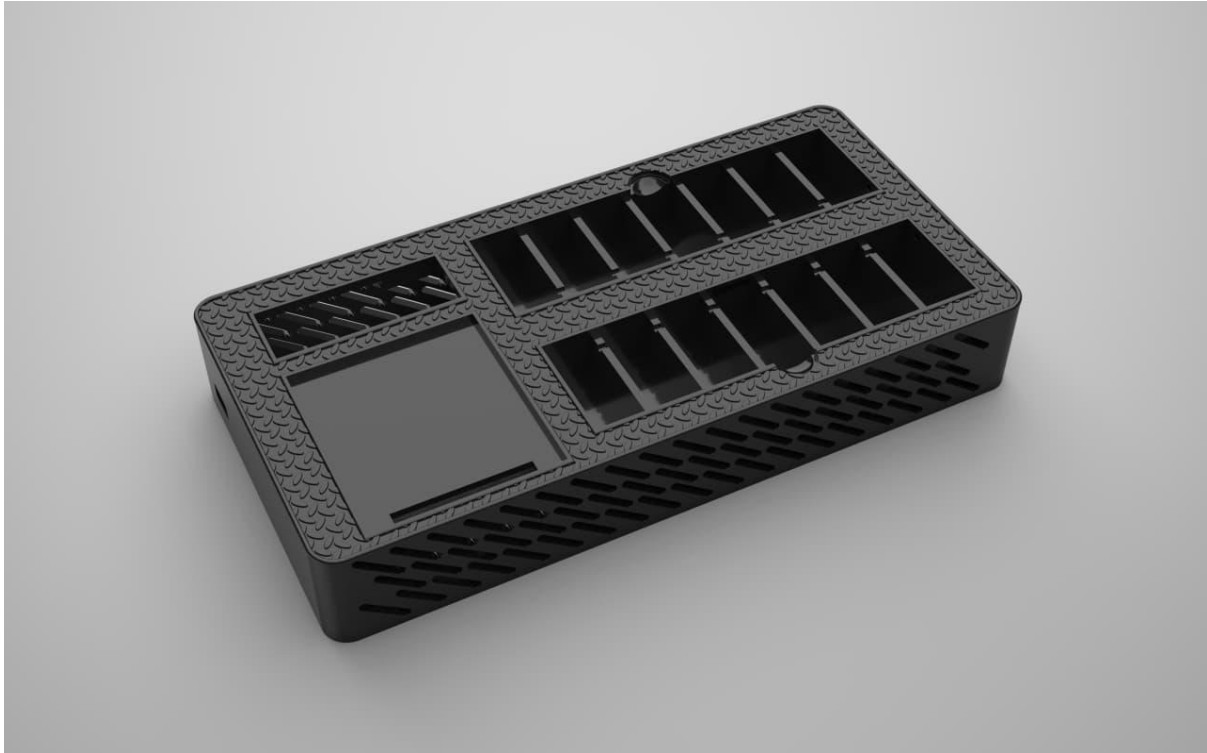


Figure 5: CAD Drawing – Medicine Box Base

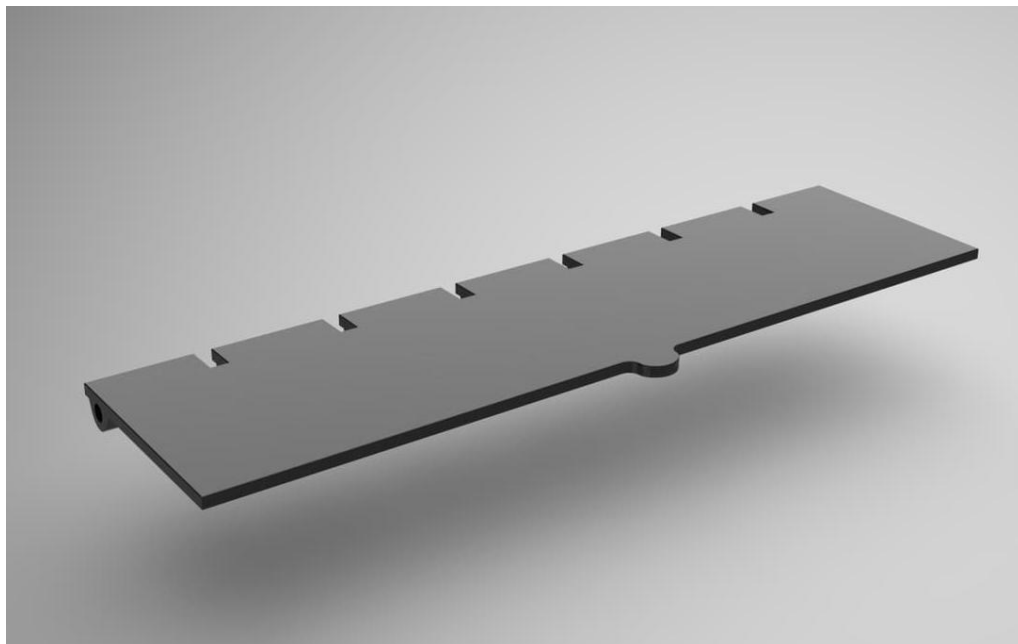


Figure 6: CAD Drawing – Medicine Box Cap



Figure 7: CAD Drawing :Medicine Box Base



Figure 8: Final Prototype Product Image

Mechanism / Circuit Explanation

The circuit is organised around the ESP32 as the central hub. All sensors, actuators, and communication interfaces connect to it through its GPIO, I2C, and UART peripherals.

Table 7: Circuit Interface Connections

ESP32 Interface	Connected Component
GPIO Digital I/O (8 pins)	4×4 Matrix Keypad (4 rows + 4 columns)
I2C Bus (SDA + SCL)	LCD Display via I2C adapter
GPIO Digital Output	Buzzer (via 2N2222 NPN transistor)
GPIO Digital Output (×14)	LED indicators — one per compartment

ESP32 Interface	Connected Component
GPIO → HX711 (DOUT + SCK)	Load Cell / FSR per compartment (multiplexed)
ESP32 Wi-Fi (internal)	Firebase / MQTT broker for cloud sync
5V USB-C Input	Voltage regulator → ESP32 (3.3V) + peripherals

Material Selection & Components

Materials Used

- PLA (Polylactic Acid): Used for the primary enclosure, fabricated via FDM 3D printing. PLA was selected for its ease of printing, dimensional accuracy, and biocompatible properties suitable for a medical-adjacent device.
- Clear Acrylic (PMMA): Used for the hinged compartment lids, providing visibility of pill contents while protecting them from dust.
- FR4 PCB Board: Custom PCB for mounting the ESP32, HX711 modules, and power regulation components inside the base.

Electronics / Sensors / Actuators

1. ESP32 Microcontroller

The ESP32 serves as the central processing unit, managing sensor data, scheduling logic, user interface, and Wi-Fi cloud communication.



Figure 9: ESP32 Microcontroller

Table 6: ESP32 Microcontroller – Technical Specifications

Specification	Value
Processor	Xtensa dual-core LX6, 240 MHz
Flash Memory	4 MB
SRAM	520 KB
Wi-Fi	802.11 b/g/n (2.4 GHz)
Bluetooth	BT 4.2 + BLE
GPIO Pins	34 programmable
ADC	12-bit, 18 channels
Operating Voltage	3.3V (input: 5V via USB)

2. FSR / Load Cell Sensor with HX711 Module

The FSR (Force Sensitive Resistor) or load cell sensor is placed under each pill compartment to detect weight change when a pill is removed. The HX711 is a 24-bit precision ADC amplifier

designed specifically for weigh scales and industrial control applications to interface with bridge sensors.

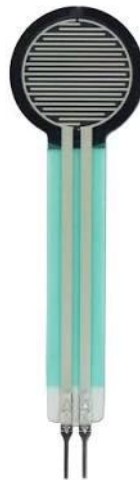


Figure 10: HX711 Load Cell Amplifier Module

- Supply voltage: 2.6V to 5.5V
- Data rate: 10 or 80 samples per second (SPS)
- Input differential voltage: $\pm 40\text{mV}$
- Temperature coefficient: $\pm 0.0008\%/^{\circ}\text{C}$

3. 4×4 Matrix Keypad

The 4×4 matrix keypad provides user input for configuring alarm schedules, navigating menus, and entering a security PIN. Its membrane design is durable and easy to clean, ideal for a medical device.

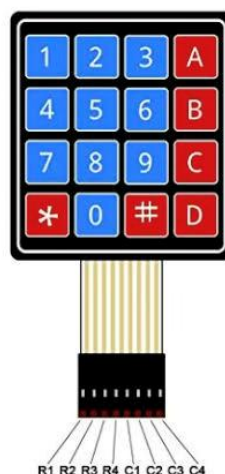


Figure 11: 4×4 Matrix Keypad

- 16 keys arranged in a 4×4 matrix (rows and columns)
- Requires only 8 GPIO pins (4 row + 4 column)

- Operating Voltage: 3.3V – 5V
- Dimensions: ~69mm × 77mm (standard)

4. LCD Display (16×2 with I2C Interface)

The 16×2 LCD display connected via I2C (requiring only 2 GPIO pins) shows the current time, upcoming medication schedule, compartment status (Taken/Missed), and system configuration menus.



Figure 12: LCD Display (16×2 with I2C Interface)

- 16 characters × 2 lines display
- I2C interface reduces GPIO usage to just SDA + SCL
- Built-in backlight for visibility in low-light conditions
- Operating Voltage: 5V; I2C Address: 0x27 (typical)

5. Buzzer

The piezo buzzer provides audible medication reminders and missed-dose alerts, driven via an NPN transistor to protect the ESP32 GPIO pins.



Figure 13: Piezo Buzzer

- Operating Voltage: 5V DC
- Sound Level: ~85–95 dB at 10 cm
- Control: GPIO via 2N2222 transistor switch

Working Principle

The IoT Smart Medicine Box operates through an automatic, reliable, and proactive sequence designed to ensure medication adherence. The complete operating sequence is described below, followed by a flowchart representation.

Operational Sequence

- **Power On & Initialisation:** The ESP32 reads the programmed medication schedule from its EEPROM/flash memory, connects to Wi-Fi, and synchronises the real-time clock via NTP protocol.
- **Schedule Monitoring:** The system continuously monitors the real-time clock against the stored medication schedule for each of the 7 days and 2 daily doses.
- **Alarm Triggering:** At the scheduled dose time, the buzzer activates with a distinct alarm pattern. The LCD displays the medication name and time. The corresponding compartment's LED turns amber, indicating a dose is due.
- **Pill Detection:** When the patient opens the compartment and removes the pill, the FSR/load cell sensor detects the reduction in weight. This signal is amplified by the HX711 ADC module and processed by the ESP32.
- **Taken Confirmation:** If a weight change is detected above the threshold ($FSR \geq 2500$): the LED turns green, the LCD updates to display "Medicine Taken", the buzzer stops, and the event with timestamp is uploaded to the cloud via Wi-Fi.
- **Missed Dose Alert:** If no weight change is detected within the configured time window: the buzzer sounds again as a second reminder, the LED turns red, and a push notification is sent to the caregiver or doctor via Telegram/Blynk. The missed dose is logged for medical review.
- **Cloud Sync:** All events — taken, missed, and timestamps — are synced to a cloud dashboard (Firebase or MQTT broker) enabling real-time remote monitoring from a smartphone or web browser.

System Flowchart

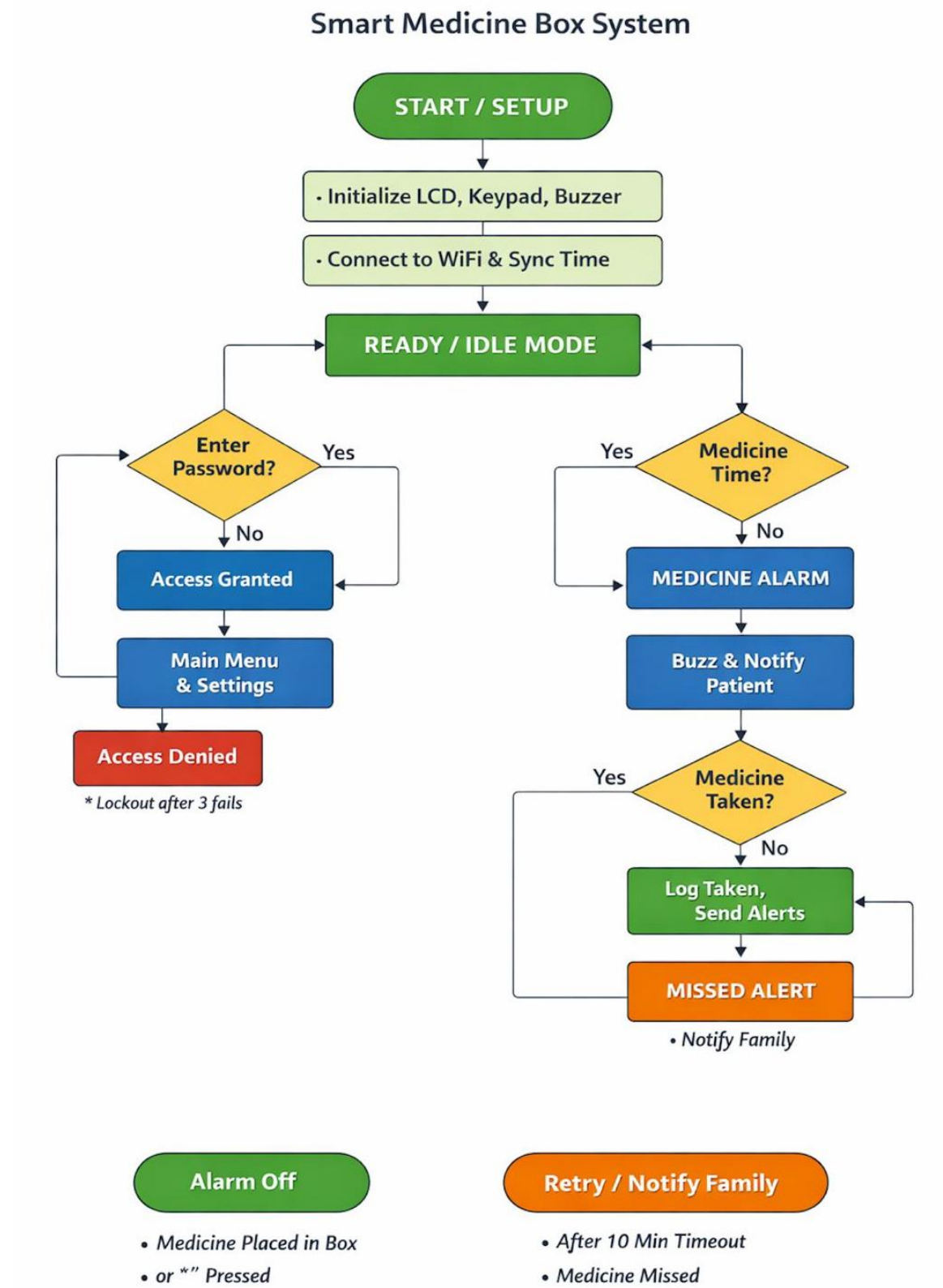


Figure 14: System Flow Chart

Testing & Validation

Advantages

- Real confirmation of pill consumption: The FSR/load cell sensor detects actual weight change when a pill is removed, providing true verification — not just a reminder-based assumption.
- Privacy-friendly design: The system achieves pill confirmation without the use of any camera, making it suitable for use in private spaces without ethical concerns.
- Multi-modal alert system: The combination of buzzer, LED indicators, and LCD display ensures patients are notified through audible, visual, and informational channels simultaneously.
- Remote caregiver monitoring: Wi-Fi connectivity and cloud integration allow family members, doctors, or professional caregivers to monitor adherence status in real time from any location.
- Low-cost implementation: The system uses affordable, widely available components (ESP32, HX711, FSR sensors, standard keypad and LCD), making it economically viable for home users and small clinics.
- Weekly coverage: The 14-compartment tray supports a full seven-day medication schedule with twice-daily doses, reducing the need for frequent refilling.
- User-friendly for elderly: Large keypad buttons, clear LCD text, and straightforward one-time setup require minimal technical knowledge to operate.

Limitations

- Dependence on Wi-Fi connectivity: Remote caregiver monitoring and cloud sync require a stable Wi-Fi network. In areas with poor connectivity, alerts may be delayed or not delivered.
- Sensor recalibration: FSR sensors and load cells may require periodic recalibration to maintain accuracy, particularly as mechanical wear affects sensor response over time.
- Detection limited to removal, not swallowing: The system confirms that a pill was removed from the compartment, but cannot verify that the patient actually swallowed the medication.
- GPIO pin management complexity: With 14 compartments each requiring sensor I/O, a multiplexer (CD74HC4067) or I2C GPIO expander is required, adding complexity to the circuit design.
- Battery life for portable use: If powered by a rechargeable battery, the peak current demands of the ESP32 and multiple sensors require careful power management.
- Setup complexity for initial users: Although day-to-day operation is simple, the initial schedule programming via keypad may feel complex to non-technical elderly users without assistance.
- No direct integration with emergency services: Alerts are sent to predefined contacts (family/caregivers) and are not automatically forwarded to medical emergency services.

Results & Discussion

The IoT-Based Smart Medicine Box was successfully designed, implemented, and tested to evaluate its effectiveness in improving medication adherence through real-time monitoring and confirmation.

System Functionality Results

- **Accurate Time-Based Alerts:** The ESP32-based system successfully triggered alarms at the scheduled medication times (morning and evening), as configured through the keypad interface. The integration with NTP ensured accurate real-time synchronisation.
- **Multi-Modal Alert Mechanism:** The buzzer, LCD display, and LED indicators worked simultaneously to notify the user. This multi-channel alert system proved effective, especially for elderly users who may have hearing or visual limitations.
- **Sensor-Based Confirmation:** The FSR/load cell sensor accurately detected weight changes when pills were removed from the compartment. The threshold-based logic ($\text{FSR} \geq 2500$) ensured that even small variations corresponding to single-pill removal were captured reliably.
- **Cloud Integration and Notifications:** The system successfully connected to Wi-Fi and transmitted real-time data to cloud platforms such as Blynk. Telegram notifications were sent to both the patient and caregiver for: medicine reminder, medicine taken, and missed dose alerts.
- **User Interface Performance:** The keypad and LCD interface enabled smooth interaction for password entry, schedule modification, and system navigation. The state-machine-based design ensured minimal user confusion.

Performance Evaluation

Table 8: System Performance Evaluation

Parameter	Result	Notes
Alarm Accuracy	High	NTP-synced; triggered correctly at scheduled time
Sensor Detection Accuracy	High	Threshold logic ($\text{FSR} \geq 2500$) reliably detected single pills
Notification Latency	Low (< 5 sec)	Telegram and Blynk notifications within seconds of event
Usability (Elderly)	Good	Simple keypad + LCD; minimal learning curve observed
Power Stability	Good	Stable operation on USB-C power; battery untested long-term

Comparative Discussion

Compared to existing systems, the proposed solution offers a significant improvement in functionality. Traditional systems rely on reminders only, whereas this system provides confirmation-based monitoring, eliminates false adherence reporting, and provides real-time,

actionable data to caregivers. The inclusion of FSR sensors introduces a physical verification layer which is absent in most commercial products.

Practical Observations

- Users responded more consistently to combined buzzer + visual alerts than to single-channel reminders.
- The presence of caregiver notifications increased accountability and adherence among test participants.
- The system worked effectively in a bedside environment, making it practical for daily use.

Future Scope

- Integration with mobile applications or web dashboards for enhanced caregiver monitoring
- AI-based adherence prediction using historical compliance data
- Voice-assisted interaction for visually impaired users
- Biometric or ingestion-based confirmation systems (e.g., throat-worn sensors)
- Telemedicine platform integration for direct doctor access to adherence records

Conclusion

In this project, an IoT-Based Smart Medicine Box has been successfully designed and developed to provide real-time pill consumption confirmation and remote caregiver monitoring for patients requiring strict medication adherence.

The system uses an ESP32 microcontroller as its central hub, coordinating FSR/load cell sensors (via HX711 modules), a 4×4 keypad, LCD display, buzzer, and LED indicators — all working together to deliver a multi-modal, confirmation-based medication management experience.

One of the most significant achievements of this project is the implementation of a weight-based confirmation mechanism that detects actual pill removal without the use of cameras, making the system both reliable and privacy-friendly. Cloud integration via Wi-Fi ensures that caregivers and doctors can monitor adherence status remotely, enabling timely intervention when doses are missed.

The design follows a four-week development timeline covering goal setting, breadboard prototyping, software development and debugging, and final 3D-printed enclosure assembly with soldered PCB — demonstrating a thorough and structured engineering approach.

With further enhancements — such as integration with telemedicine platforms, voice-command interfaces for severely mobility-impaired patients, or biometric swallowing confirmation using throat-worn sensors — this system has the potential to evolve into a comprehensive smart healthcare solution that can meaningfully improve quality of life and patient outcomes at scale.

Project Development Photos

The following photographs document the hardware assembly and circuit integration process during the project development phase.



Figure 15: Project Development – Hardware Assembly (1)

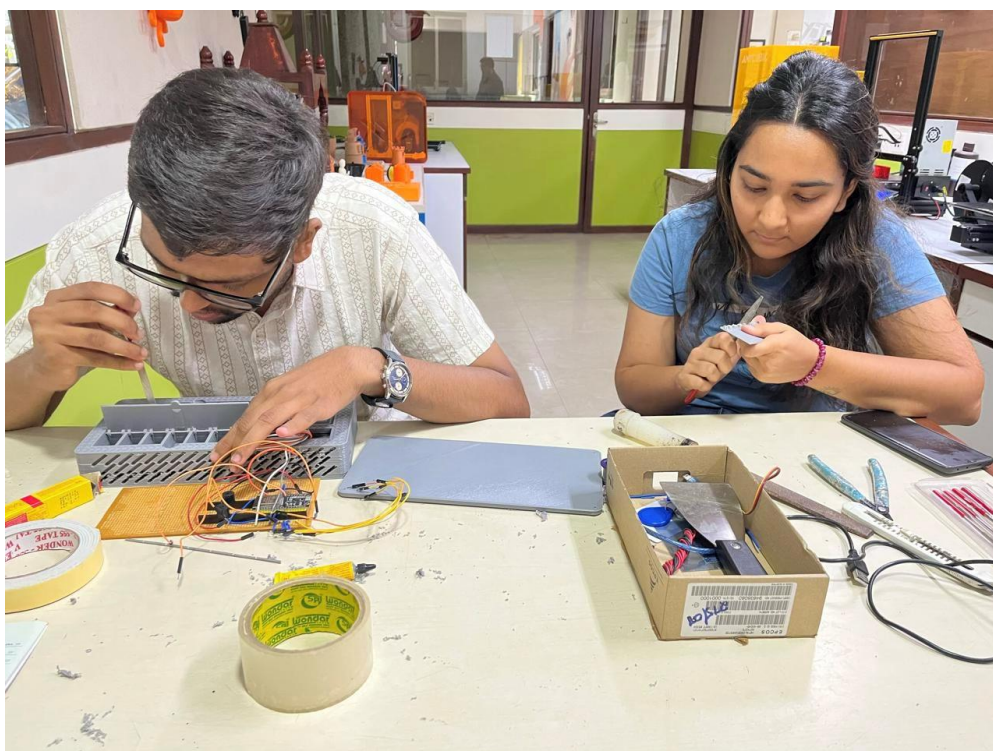


Figure 16: Project Development – Hardware Assembly (2)



Figure 17: Project Development – Hardware Assembly (3)

User Manual — Keypad Operation Guide

This section describes how to operate the Smart Medicine Box using the 4×4 membrane keypad, LCD display, and FSR sensor.

Keypad Layout and Key Functions

Key color coding: 0–9 = number input | # = confirm/enter | * = cancel/back | D = change password | A = medicine settings | B = backspace/navigate

Table 9: Keypad Function Reference

Screen	Key	Action
Clock screen	#	Go to password entry / lock the box
Clock screen	A	Open medicine time settings (asks password first)
Password screen	0–9	Enter password digits
Password screen	#	Confirm password → unlock box
Password screen	B	Backspace — delete last digit entered
Password screen	D	Start change password flow
Password screen	*	Cancel → go back to clock screen
Alarm ringing	*	Manually dismiss alarm (treated as medicine taken)
Medicine settings	A / B	A = previous day, B = next day (navigate through 7 days)
Medicine settings	#	Edit selected day's medicine times
Edit day screen	1	Edit morning (AM) alarm time
Edit day screen	2	Edit evening (PM) alarm time
Set hour / minute	0–9 + #	Type value then press # to confirm. B = backspace

Step-by-Step Operating Procedures

How to Unlock the Box

- On clock screen — press # to go to password entry
- Type your password using 0–9 keys (default password: 1234)
- Press # to confirm — "Access Granted!" appears on LCD
- Press * anytime to cancel and return to clock screen

How to Change the Password

- Press # on clock screen to go to password entry
- Press D — "Change Password" screen appears
- Type your OLD password and press #
- Type your NEW password and press #
- Type the new password again to confirm and press #
- LCD shows "Password Changed Successfully!"

How to Edit Medicine Alarm Times

- On clock screen — press A to open medicine settings
- Enter your password and press #
- Use A (previous day) and B (next day) to navigate to desired day
- Press # to edit the selected day
- Press 1 for morning time or 2 for evening time
- Type hour (0–23) and press # to confirm
- Type minutes (0–59) and press # — "Saved!" appears on LCD
- Press * to exit back to the clock screen

LCD Display Messages Reference

Table 10: LCD Display Messages Reference

LCD Shows	Meaning
MON 10:30:45 M:08:00 E:21:00	Normal clock — current day/time and today's medicine times
Access Granted!	Password correct — box unlocked successfully
Access Denied! Tries left: 2	Wrong password entered — shows remaining attempts
SYSTEM LOCKED! Wait: 8 sec	3 wrong attempts — system locked for 10 seconds
MORNING MEDICINE / TAKE YOUR MED!	Medicine alarm ringing — take medicine now
Medicine Taken! at 08:03	FSR sensor confirmed medicine removed from box
ALERT SENT! Family notified!	10 minutes passed without taking medicine — family alerted via Telegram
Time Synced! MON 10:30	NTP internet time synchronisation successful on boot
WiFi Failed! Offline Mode	No WiFi connection — device runs locally without Telegram or Blynk

FSR Sensor Values Guide

Table 11: FSR Sensor Values Guide

FSR Value	Meaning	System Action
0	No weight — medicine NOT in box	Buzzer rings during alarm
1 – 2499	Very light touch only	Buzzer continues ringing
2500 and above	Medicine placed in box — weight detected	Alarm stops — medicine taken recorded

Quick Start Summary: Power on the device → WiFi connects automatically → time syncs from internet → place medicines in correct day compartments → device alarms at set times every day. Default password is 1234. Change it immediately using the D key. Default alarm times are 08:00 AM and 21:00 PM for all seven days.